

Code-Layout, Readability and Reusability

Date	11 July 2026
Team ID	SWTID-2026-1298
Project Name	Smart Lender
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Consistent formatting maintained across all source files.
2	Proper File Structure	Files and folders are logically organized	Yes	Project organized into templates, static, models, datasets, and application files.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Descriptive naming convention followed.
4	Function / Method Names	Functions are descriptively named	Yes	Function names clearly indicate their purpose.
5	Code Comments	Inline and block comments explain logic	Partial	Comments included for important functions and processing steps.
6	Modular Design	Code is split into reusable functions/modules	Yes	Independent modules created for preprocessing, prediction, and web application.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Common logic reused through functions to minimize duplication.
8	Error Handling	Exceptions and errors are handled gracefully	Partial	Basic input validation implemented; advanced exception handling can be enhanced.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	Authentication Module	Python, Flask	Login and session management	High
2	Data Preprocessing Module	Python, Pandas	Data cleaning and feature preparation	High
3	Prediction Engine	Python, XGBoost	Loan approval prediction	High
4	User Interface Templates	HTML, Bootstrap	Multiple web pages	Medium
5	Database Access Module	SQLite, Flask	Applicant records and prediction history	High

Overall Code Quality Assessment:

Aspect	Rating (1–5)	Comments
Code Layout & Structure	5	Project follows a clean folder hierarchy and modular organization.
Readability	5	Meaningful naming conventions and proper formatting improve code clarity.
Reusability	5	Common functionalities are implemented as reusable modules.
Documentation / Comments	4	Important sections are documented; additional comments can be added if required.
Overall Score	4.8 / 5	The codebase is well-structured, maintainable, and suitable for future enhancements.